

Lagrangian Variable Cosmology final-v1.1: An Exact Topological Reconstruction of a Localised θ -Coordinate Modulation in Late-Time Expansion History

Working Paper

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Abstract

This paper is the latest in a working-paper series (v1–v16 phenomenological line, v0.1–v0.3 theoretical reconstruction; all archived at Zenodo, see References). A localised modulation of the late-time expansion rate has been identified in successive fits to combined Pantheon+, DESI DR1+DR2, BOSS DR12, eBOSS DR16, Planck 2018 distance priors, and SH0ES data; in the logarithmic-time coordinate $\theta = \ln(1+z)$, free-fit profiles cluster around a centre $\theta_c = \ln \varphi$ with φ the golden ratio. The phenomenological status is documented in v0.1–v0.3. The present paper attempts an exact topological reconstruction of this profile: starting from the observation that the data prefer a narrow, near-symmetric feature at $\theta_c = \ln \varphi$, we ask whether such a profile can arise as the classical kink-sector solution of a single-scalar field theory. The half-twist sine-Gordon potential $V(\vartheta) = m^2[1 - \cos(\vartheta/2)]$ admits the kink $\vartheta_{\text{kink}}(\theta) = 8 \arctan(\exp[(m/2)(\theta - \theta_c)])$, and the identification $v(\theta) \equiv T(\theta) - 1 = \varepsilon^2 \sin^2(\vartheta_{\text{kink}}/4)$ yields $v(\theta) = A \operatorname{sech}^2((\theta - \theta_c)/w)$ with $A \equiv \varepsilon^2$ and $w \equiv 2/m$. The induced equation in v is the BPS-saturated bounce of a cubic effective potential $V_{\text{eff}}(v) = -(2/w^2)v^2(1 - v/A)$. Two structural relations follow: linearisation near $v = 0$ fixes $w = A/\ln \varphi$ once $\theta_c = \ln \varphi$ is imposed, and the condition $\ln \varphi = \varphi \cdot \theta_{\text{eq}}(\Omega_m)$ is equivalent to $\Omega_m = 1/(1 + e^{3 \ln \varphi / \varphi}) = 0.290653$. On the PP-excluded data set ($N = 37$) the full-lock construction returns $\chi^2 = 56.134$ at $k = 2$, against $\chi^2 = 78.962$ at $k = 3$ for Λ CDM, giving $\Delta\text{BIC} = -26.44$. On the full PP-included data set ($N = 1738$) the same locks return $\chi^2 = 1601.999$ at $k = 2$, against $\chi^2 = 1624.848$ at $k = 3$ for Λ CDM, giving $\Delta\text{BIC} = -30.31$; the locks survive the $47\times$ increase in measurement count and the introduction of the supernova distance ladder. Locking four of six modulation parameters costs only $\Delta\chi^2 = +1.00$ relative to the unconstrained six-parameter fit on the same data, indicating that the locks point essentially at the data preference within the resolving power of the present datasets. The reconstruction is offered as an exact existence result, not as a fundamental theory: $\theta_c = \ln \varphi$ and the value of A are imposed inputs, not derived from the action; the selection principle that would fix them remains open.

1 Introduction

A localised feature in the late-time cosmological expansion rate has been identified across successive working papers (v1–v16) on the joint dataset of Pantheon+ unbinned supernovae [6], DESI DR1 and DR2 BAO [5], BOSS DR12 [9], eBOSS DR16 [10], Planck 2018 compressed distance priors [8], and SH0ES H_0 [7]. The feature takes the form of a multiplicative modulation $T(z) = H(z)/H_{\Lambda\text{CDM}}(z)$ that returns to unity at $z = 0$ and at high redshift and exhibits

a narrow excursion centred near $z \approx 0.62$. In the logarithmic-time coordinate $\theta = \ln(1+z)$, free-fit profiles centre on $\theta_c = 0.4812 = \ln \varphi$ to four decimal places, with $\varphi = (1 + \sqrt{5})/2$ the golden ratio.

The phenomenological status of the feature, its coordinate dependence, and its robustness under bootstrap and dataset reduction are documented in v0.1–v0.3 of this series [1–3], and the precursor combined-fit identification is recorded in v16 [4]. The present paper does not extend the phenomenology; it attempts a different question. Granting the empirical fact that the data prefer a narrow, near-symmetric excursion at $\theta_c = \ln \varphi$ with width $w \approx 0.110$ (v0.3 free fit), *can such a profile arise as the classical kink-sector solution of a single-scalar field theory?*

This question admits a clean affirmative answer. The half-twist sine–Gordon potential $V(\vartheta) = m^2[1 - \cos(\vartheta/2)]$, distinguished from the standard sine–Gordon potential by its 4π rather than 2π period, admits a static one-soliton kink whose squared quarter-angle is exactly $\text{sech}^2((\theta - \theta_c)/w)$. Identifying the modulation deviation amplitude with $\varepsilon^2 \sin^2(\vartheta_{\text{kink}}/4)$ produces the observed profile up to the choice of ε^2 and θ_c . The induced equation in the deviation amplitude v is the BPS-saturated bounce of a cubic effective potential, and two structural relations follow without further input.

The paper is organised in two parts. Part I (Sections 2–3) records the empirical structure: the localised modulation in θ -coordinates (Section 2), and the response of the lock candidates under a $47\times$ increase in dataset size (Section 3). Part II (Sections 4–5) records the reconstruction: the half-twist sine–Gordon kink and its sech^2 image (Section 4), and the two structural relations that follow (Section 5). Limitations are stated in Section 6 and the outlook in Section 7.

The discipline of this paper is to keep the phenomenology and the reconstruction separate. The reconstruction is an inverse program: given the observed profile, an exact field-theoretic sector that admits it as a classical kink solution is exhibited. No claim is attached to the choice of action beyond the consistency of its kink solution with the previously fitted modulation. In particular, neither $\theta_c = \ln \varphi$ nor the value of A is selected by any internal principle of the construction; both are imposed inputs.

2 Phenomenological observations

The model class fixed since v0.1 is a multiplicative modulation of the Λ CDM expansion rate,

$$H(z) = H_{\Lambda\text{CDM}}(z; H_0, \Omega_m) \cdot T(z), \quad (1)$$

with $T(z) \rightarrow 1$ at $z = 0$ and at high redshift. Two single-bump parametrisations of the modulation factor have been compared across v0.1–v0.3:

$$\text{F1, } z\text{-Gaussian: } T(z) = 1 + A \exp[-((z - z_c)/w)^2], \quad (2)$$

$$\text{F2, } \theta\text{-Gaussian: } T(z) = 1 + A \exp[-((\theta - \theta_c)/w)^2], \quad \theta = \ln(1+z). \quad (3)$$

The free-fit results recorded in v0.3 [3] on the PP-included data set ($N = 1738$, comprising Pantheon+ unbinned $N = 1701$, DESI DR2 BAO $N = 13$, DESI DR1 BAO $N = 11$, BOSS DR12 $N = 2$, eBOSS DR16 $N = 7$, Planck 2018 distance priors $N = 3$, and SH0ES H_0 $N = 1$) are summarised below:

- F1 free fit: $\chi^2 = 1601.189$ at $z_c = 0.6180$, $w = 0.171$. The position equals $1/\varphi$ to four decimal places.
- F2 free fit: $\chi^2 = 1600.997$ at $\theta_c = 0.4812$, $w = 0.110$. The position equals $\ln \varphi$ to four decimal places.

Three observations follow from the v0.3 free-fit results and the bootstrap and profile-likelihood diagnostics performed there.

Observation 1: position locking. Across both parametrisations and across PP-included and PP-excluded subsets, the free-fit centre returns to a natural-constant value (in F2: $\ln \varphi$) within the precision of the fit. On PP-included data the recovery is to four decimal places.

Observation 2: width preference. The F2 free-fit width $w \approx 0.110$ on PP-included data lies -36% below the v0.1 candidate $e/(5\pi) = 0.1731$. The deviation is robust across subsets (PP-excluded F2: $w = 0.116$). The numerical value $w \approx 0.110$ is close to $(\ln \varphi)^3 = 0.11143$ and to $2/\varphi^6 = 0.11146$; the latter two differ from each other by 0.02% and are not distinguishable in the present data.

Observation 3: profile family degeneracy. Single-bump functional families (Gaussian in z , Gaussian in θ , skewed, narrow) all fit the data within $\Delta\chi^2 < 1$ of each other in v0.2 mock validations [2]. Double-bump and monotonic alternatives are excluded at $\Delta\chi^2 > 4$. The single-bump localisation is robust; the specific functional shape within that class is not selected by data alone.

The position lock and the narrow-width preference, taken together, define a constraint surface in the (θ_c, w, A) modulation space. The remainder of this paper records what happens when the constraint is taken seriously: first, when it is tested under enlargement of the dataset (Section 3), and second, when a field-theoretic sector is sought whose classical kink coincides with the observed profile (Section 4).

3 Structural constraints under dataset scale-up

The v0.3 free fits identified $\theta_c = \ln \varphi$ to four decimal places and a width near $(\ln \varphi)^3$. Treating these as fixed inputs — not as derived predictions — one can ask whether the resulting locked construction survives an enlargement of the dataset, and to what extent the locks coincide with the data preference.

3.1 Lock specification

We adopt the following lock pattern, taken from the v0.3 free-fit numerical values and the natural-constant identifications they suggest:

$$\theta_c = \ln \varphi, \quad A = (\ln \varphi)^4, \quad w = (\ln \varphi)^3, \quad \Omega_m = \frac{1}{1 + \exp(3 \ln \varphi / \varphi)}. \quad (4)$$

The construction has two free parameters, H_0 and $\Omega_b h^2$. The motivation for the algebraic forms of w and Ω_m in terms of $\theta_c = \ln \varphi$ is given in Section 5 and emerges from the reconstruction of Section 4.

3.2 PP-excluded fit ($N = 37$)

On the PP-excluded data set — DESI DR1 BAO, DESI DR2 BAO, BOSS DR12, eBOSS DR16, Planck 2018 distance priors, and SH0ES H_0 , totalling $N = 37$ measurements — the locked construction returns $\chi^2 = 56.134$ at $k = 2$, against $\chi^2 = 78.962$ at $k = 3$ for Λ CDM, giving $\Delta\text{BIC} = -26.44$.

The candidate $\Omega_m = 1/(2 + \varphi) = 0.27639$, which would correspond to the algebraic identity $\rho_m/\rho_\Lambda = \varphi$ at $\theta = \ln \varphi$, returns $\chi^2 = 81.83$ at the same locks ($\Delta\chi^2 = +25.7$ relative to the free- Ω_m reference, and above the Λ CDM value). The candidate Eq. (4) and the candidate $1/(2 + \varphi)$ are therefore distinguished by the present data at large significance; (4) is recorded as compatible and $1/(2 + \varphi)$ as not.

3.3 PP-included fit ($N = 1738$): $47\times$ scale-up

The same lock pattern Eq. (4) was fitted to the full PP-included data set, increasing the measurement count from $N = 37$ to $N = 1738$, a factor of 47. The likelihood pipeline is the v0.1 reference implementation [1] as reproduced in v0.3 [3]. No re-tuning of the locks is performed; the values of θ_c , A , w , and Ω_m in Eq. (4) are held fixed.

The result is summarised in Table 1. The locked construction returns $\chi^2 = 1601.999$ at $k = 2$, against $\chi^2 = 1624.848$ at $k = 3$ for Λ CDM, giving $\Delta\text{BIC} = -30.31$. The lock survives the introduction of the full Pantheon+ unbinned likelihood, and ΔBIC deepens by -3.87 relative to the PP-excluded value.

Table 1: Scale-up of the locked construction from PP-excluded ($N = 37$) to PP-included ($N = 1738$). The locks of Eq. (4) are held fixed in both rows. ΔBIC is computed against the Λ CDM free fit on the same data set with the same $k = 3$ counting (free H_0 , Ω_m , $\Omega_b h^2$).

<i>Data set</i>	<i>N</i>	χ^2 (locked, $k = 2$)	χ^2 (Λ CDM, $k = 3$)	ΔBIC
PP-excluded	37	56.134	78.962	-26.44
PP-included	1738	1601.999	1624.848	-30.31

3.4 Locks versus the unconstrained best fit

A separate question is how far the locks Eq. (4) differ from the global best fit on the same data when the modulation parameters are not fixed. Successive relaxations on the PP-included data set ($N = 1738$) give:

- Locked construction (Eq. (4), $k = 2$): $\chi^2 = 1601.999$.
- Same locks on θ_c, A, w but Ω_m free ($k = 3$): $\chi^2 = 1601.95$.
- Same lock on θ_c but A and Ω_m free, w tied to A via Eq. (12) ($k = 4$): $\chi^2 = 1601.07$.
- All six parameters free (full F2 free fit, $k = 6$, from v0.3): $\chi^2 = 1600.997$.

The full lock pattern Eq. (4) costs only $\Delta\chi^2 = +1.00$ relative to the unconstrained six-parameter fit, while removing four of the six modulation parameters. Within the resolving power of the present datasets, the locks point essentially at the data preference; the data-preferred minimum is not visibly distinguishable from the locked configuration.

3.5 Comments on the structural pattern

Three features of the scale-up are recorded without further commentary:

- The locks were fixed using the v0.1 phenomenology and the v0.3 free-fit width identification. Their survival on the $N = 1738$ data set is a check, not a fit.
- The improvement of ΔBIC from -26.44 to -30.31 reflects both the additional information content of Pantheon+ and the residual cost of locking four parameters in the larger data set; the latter is small ($\Delta\chi^2 = +1.00$).
- Among neighbouring natural-constant candidates for the ratio $R \equiv \theta_c/\theta_{\text{eq}}$, an internal scan with all other locks held returns $\Delta\chi^2 < 1$ only in the interval $R \in [1.61, 1.62]$ on the PP-excluded data set, with $R = \varphi$ at the minimum. Adjacent candidates such as $R = 8/5 = 1.600$ ($\Delta\chi^2 = +0.49$), $R = \pi/2 \approx 1.571$ ($\Delta\chi^2 = +3.78$), and $R = \sqrt{e} \approx 1.6487$ ($\Delta\chi^2 = +1.58$) are within or close to the 1σ band but do not improve on $R = \varphi$. The data therefore distinguish φ from its closest natural-constant neighbours at the $\sim 0.5\%$ level on the present scan.

4 Exact topological reconstruction

The phenomenology of Sections 2–3 fixes a target profile: a localised, near-symmetric excursion in the deviation amplitude $v(\theta) \equiv T(\theta) - 1$, centred on $\theta_c = \ln \varphi$, with width w near $(\ln \varphi)^3$, and asymptoting to zero on both sides. We now ask whether such a profile arises as the classical kink-sector solution of a field theory with a single real scalar.

4.1 Action and half-twist sine–Gordon potential

The action considered is that of a single real scalar ϑ minimally coupled to gravity:

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{2\kappa} + \mathcal{L}_{\text{matter}} + \mathcal{L}_\vartheta \right], \quad (5)$$

$$\mathcal{L}_\vartheta = \frac{1}{2} g^{\mu\nu} (\partial_\mu \vartheta)(\partial_\nu \vartheta) - V(\vartheta), \quad V(\vartheta) = m^2 [1 - \cos(\vartheta/2)]. \quad (6)$$

The potential $V(\vartheta)$ is periodic with period 4π , distinguishing it from the standard sine–Gordon potential $m^2(1 - \cos \vartheta)$ of period 2π . The choice of period is the central reconstruction-side input of this section; its empirical justification rests on the agreement of the resulting kink profile with the phenomenology.

4.2 Kink solution and sech² image

On the static one-soliton sector the equation of motion $\vartheta_{\theta\theta} = (m^2/2) \sin(\vartheta/2)$ admits the kink

$$\vartheta_{\text{kink}}(\theta) = 8 \arctan(\exp[(m/2)(\theta - \theta_c)]), \quad (7)$$

winding from $\vartheta = 0$ at $\theta \rightarrow -\infty$ through $\vartheta = 2\pi$ at $\theta = \theta_c$ to $\vartheta = 4\pi$ at $\theta \rightarrow +\infty$. Direct computation gives $\sin(\vartheta_{\text{kink}}/4) = \text{sech}[(m/2)(\theta - \theta_c)]$ and $\cos(\vartheta_{\text{kink}}/4) = -\tanh[(m/2)(\theta - \theta_c)]$, so that

$$\sin^2(\vartheta_{\text{kink}}(\theta)/4) = \text{sech}^2[(m/2)(\theta - \theta_c)]. \quad (8)$$

Identifying the modulation deviation amplitude with the squared sine of the compact-phase quarter-angle along the kink,

$$v(\theta) \equiv T(\theta) - 1 = \varepsilon^2 \sin^2(\vartheta_{\text{kink}}(\theta)/4), \quad (9)$$

yields $v(\theta) = A \text{sech}^2((\theta - \theta_c)/w)$ with $A \equiv \varepsilon^2$ and $w \equiv 2/m$. Equation (8) is an algebraic consequence of (7) and is verified numerically to machine precision; no approximation is involved at this stage.

4.3 Effective equation of motion and BPS structure

Working directly with $v(\theta)$, the classical equation of motion induced by the substitution (9) is autonomous in v and takes the form $v_{\theta\theta} + V'_{\text{eff}}(v) = 0$, with the cubic effective potential

$$V_{\text{eff}}(v) = -\frac{2}{w^2} v^2 \left(1 - \frac{v}{A} \right). \quad (10)$$

The sech² profile is the unique trajectory satisfying $v(\theta \rightarrow \pm\infty) = 0$, $v(\theta_c) = A$, $v_\theta(\theta_c) = 0$, with the energy integral $\frac{1}{2} v_\theta^2 + V_{\text{eff}}(v) = 0$ along the entire trajectory; equivalently

$$V_{\text{eff}}(v(\theta)) = -\frac{1}{2} v_\theta(\theta)^2 \quad (\text{BPS-saturated bounce}). \quad (11)$$

The identities (10) and (11) hold pointwise in θ . Numerical checks at $\theta \in \{0.30, 0.40, 0.481, 0.50, 0.60, 0.70\}$ return absolute residuals below 10^{-16} for $v_{\theta\theta} + V'_{\text{eff}}(v)$ and below 10^{-17} for $V_{\text{eff}}(v) + \frac{1}{2} v_\theta^2$. The

integral over the bounce returns $\int v_\theta^2 d\theta = 16 A^2/(15 w)$, in agreement with the closed-form computation using $\int \text{sech}^4 u du = 4/3$ and $\int \text{sech}^6 u du = 16/15$.

4.4 Cosmic friction and the percent-level deviation

Hubble friction in the cosmological equation of motion does not preserve (8) as an exact solution. Numerical integration of the friction-equipped equation $v_{\theta\theta} + (3w_m/2 - 3)v_\theta + V'_{\text{eff}}(v) = 0$, started from $v(\theta_c) = A$, $v_\theta(\theta_c) = 0$ with $w_m(\theta) = \Omega_m e^{3\theta}/(\Omega_m e^{3\theta} + 1 - \Omega_m)$, deviates from sech^2 with a fractional error of 0.13% of A at $|\theta - \theta_c| = 0.05$ (one half-width), 1.98% at $|\theta - \theta_c| = 0.10$ (one width), and rising sharply outside the bounce window. The integrated friction over the symmetric profile vanishes by parity. The sech^2 form is therefore an exact solution of the autonomous v -EOM (Eq. 10) and an approximate solution, controlled at the percent level, of the cosmologically friction-equipped EOM within the bounce window.

5 Structural relations

Two algebraic relations connect parameters of the construction once $\theta_c = \ln \varphi$ is taken as input. They are recorded here as consequences of the reconstruction; neither is derived from the action of Eq. (6) as a first-principles selection.

5.1 Width relation $w = A/\ln \varphi$

Linearising $V'_{\text{eff}}(v)$ near $v = 0$ retains the leading term $V'_{\text{eff}}(v) \approx -(4/w^2)v$, so that the asymptotic decay rate of $v(\theta)$ is $2/w$. Imposing the dimensionless rescaling $x = v/A$ renders V_{eff} in the form $V_{\text{eff}}(x) = -2(\ln \varphi)^2 x^2(x - 1)$ if and only if

$$w = A/\ln \varphi. \quad (12)$$

Equation (12) is one constraint relating two of the three modulation parameters (θ_c, w, A) ; it does not by itself pin any of them. The choice of normalisation $2(\ln \varphi)^2$ is the input that promotes (12) from a definition to a constraint.

The linearisation argument reproduces the v0.3 free-fit width discovery [3]: on PP-included data the F2 free fit returns a best-fit width $w \approx 0.110$, deviating from the v0.1 lock $w = e/(5\pi) = 0.1731$ by -36% . The relation (12) with $A = (\ln \varphi)^4$ predicts $w = (\ln \varphi)^3 = 0.1114$, in agreement with the v0.3 free-fit value to within 1%. The width relation is therefore the structural consequence of the cubic potential (10) that matches the data-side finding.

Direct test on the PP-excluded data with $\theta_c = \ln \varphi$ imposed and (12) imposed, leaving $(H_0, \Omega_m, \Omega_b h^2, A)$ free ($k = 4$), returns $A_{\text{fit}} = 0.05469$ and $w_{\text{derived}} = 0.11366$ with $\chi^2 = 56.113$. The reference $(\ln \varphi)^k$ candidate values are $A = (\ln \varphi)^4 = 0.05362$ and $w = (\ln \varphi)^3 = 0.11143$; the ratios $A_{\text{fit}}/(\ln \varphi)^4$ and $w_{\text{derived}}/(\ln \varphi)^3$ coincide at 1.020 to four significant figures, consistent with (12). Locking in addition $A = (\ln \varphi)^4$ (so that $w = (\ln \varphi)^3$) returns $\chi^2 = 56.134$ at $k = 3$.

5.2 Recursive relation $\ln \varphi = \varphi \cdot \theta_{\text{eq}}(\Omega_m)$

Define the matter- Λ equality epoch in the $\theta = \ln(1+z)$ coordinate by $\rho_m(\theta_{\text{eq}}) = \rho_\Lambda(\theta_{\text{eq}})$, giving $\theta_{\text{eq}}(\Omega_m) = (1/3) \ln[(1 - \Omega_m)/\Omega_m]$. Imposing the algebraic identity

$$\ln \varphi = \varphi \cdot \theta_{\text{eq}}(\Omega_m), \quad (13)$$

i.e. $\theta_{\text{eq}} = \ln \varphi / \varphi$, is equivalent to

$$\Omega_m = \frac{1}{1 + \exp(3 \ln \varphi / \varphi)} = 0.290653. \quad (14)$$

Equation (13) is a self-referential identity: the modulation centre $\theta_c = \ln \varphi$ and the matter- Λ equality $\theta_{\text{eq}} = \ln \varphi / \varphi$ stand in the ratio φ to one. It is recorded as an algebraic relation; no derivation of (13) from the action of Section 4 is offered.

Direct test on the PP-excluded data with $\theta_c = \ln \varphi$, $A = (\ln \varphi)^4$, $w = (\ln \varphi)^3$, and Ω_m locked to (14) leaves only $(H_0, \Omega_b h^2)$ free ($k = 2$) and returns $\chi^2 = 56.1344$ against the free- Ω_m reference $\chi^2 = 56.1340$ ($\Omega_{m,\text{fit}} = 0.290591$). The $\Delta\chi^2 = +0.0005$ incurred by the lock is below the precision of the fit; $\Delta\text{BIC} = -3.61$ due to the removed parameter.

5.3 Combined fit table

Table 2 collects the goodness-of-fit values on the PP-excluded data set. The full lock pattern Eq. (4) corresponds to row 3, with Ω_m locked by Eq. (14).

Table 2: Goodness of fit on the PP-excluded data set ($N = 37$). k denotes the number of continuous parameters fitted. $\text{BIC} = \chi^2 + k \ln N$. All entries other than the ΛCDM reference share the action Eq. (6) and the locks $\theta_c = \ln \varphi$, $A = (\ln \varphi)^4$, $w = (\ln \varphi)^3$; they differ only in whether Ω_m is fitted, locked by Eq. (14), or locked to $1/(2 + \varphi)$.

<i>Model</i>	<i>k</i>	χ^2	<i>BIC</i>	ΔBIC vs ΛCDM
ΛCDM (free H_0 , Ω_m , $\Omega_b h^2$)	3	78.962	89.795	—
$(\ln \varphi)^k$ locks; Ω_m free	3	56.134	66.967	−22.83
Full lock (Eq. 4); Ω_m locked by Eq. (14)	2	56.134	63.357	−26.44
Same with $\Omega_m = 1/(2 + \varphi)$ instead of (14)	2	81.83	89.05	−0.74

6 Limitations

The construction is an inverse reconstruction, not a first-principles derivation. Several specific limitations are stated in plain form.

Imposed inputs. The values $\theta_c = \ln \varphi$ and $A = (\ln \varphi)^4$ are imposed; neither is selected by any internal principle of the action of Section 4. The natural-constant identification $\theta_c = \ln \varphi$ is taken from the v0.3 free fit [3]; the exponent $k = 4$ in $A = (\ln \varphi)^k$ is selected on the present data only by the χ^2 minimum across integer powers. Candidates $k = 3$ and $k = 5$, with the same Lagrangian form and $\theta_c = \ln \varphi$ imposed and $w = A/\ln \varphi$, return χ^2 larger by $\Delta\chi^2 = +17.6$ and $+6.6$ respectively.

Indistinguishable candidates. The candidate $A = (\ln \varphi)^4 = 0.0536$ and the alternative $A = 1/\varphi^6 = 0.0557$ differ by $\Delta\chi^2 = 0.001$ on PP-excluded data and by $\Delta\chi^2 = 0.04$ on PP-included data, both within the 1σ region. The numerical near-coincidence $(\ln \varphi)^3 - 2/\varphi^6 \approx -2.4 \times 10^{-5}$ propagates this ambiguity to the width: $(\ln \varphi)^3$ and $2/\varphi^6$ are likewise indistinguishable in the present fit. The PP-included data slightly prefers $1/\varphi^6$ at the $\Delta\chi^2 = 0.04$ level, but the difference is below the resolving power of the dataset.

No selection principle for φ . The reconstruction does not derive φ from the action of Section 4. A single-scalar 1+0D Lagrangian does not contain enough structure to select natural-constant ratios. The agreement of the data with $\theta_c = \ln \varphi$ to four decimal places, and the R -scan exclusion of neighbouring candidates such as $\pi/2$ and $\sqrt{e}$ at the 1σ – 2σ level, are recorded as empirical facts of the present datasets; they are not predictions of the action.

Effective potential not globally well-posed. The cubic effective potential $V_{\text{eff}}(v)$ of Eq. (10) is unbounded below for $|v| > A$ and is therefore not a candidate for a globally well-posed potential in v . The construction is consistent in this respect because the dynamical variable is the compact phase ϑ , not v : the potential $V(\vartheta)$ of Eq. (6) is bounded and periodic, and Eq. (10) is a derived effective description on the kink trajectory only.

Reconstruction nature. The reconstruction is offered as an exact existence result: an exact field-theoretic sector whose classical kink solution coincides with the empirically preferred profile. It is not offered as a fundamental description of the modulation. Whether the kink

sector reflects underlying physics, or merely captures the symmetry structure of a localised feature whose origin is different, cannot be decided within the present framework.

7 Outlook

The next decisive external input is DESI DR3. The discriminants that DR3 may resolve are:

- Whether the modulation feature persists at the present significance.
- Whether $A = (\ln \varphi)^4$ or $A = 1/\varphi^6$ (or neither) is selected.
- Whether the R -scan band $R \in [1.61, 1.62]$ tightens around $R = \varphi$ or shifts.
- Whether $\Delta\text{BIC} \approx -30$ on the full likelihood is maintained.

The reconstruction of Section 4 is a 1+0D scalar sector and does not address why the data prefer φ specifically. Routes to a selection principle would extend the framework: multi-field embeddings, topological QFT structures, or quantum-symmetry origins of the golden ratio. None of these is pursued in the present paper, and none is required for the existence result of Section 4.

8 Conclusion

A localised modulation of the late-time expansion rate, identified phenomenologically across the v1–v16 line and consolidated in v0.1–v0.3, has been reconstructed as the exact classical kink-sector solution of a single-scalar half-twist sine–Gordon Lagrangian. The squared sine of the kink quarter-angle yields a sech^2 profile in the deviation amplitude, and the induced cubic effective potential admits this profile as a BPS-saturated bounce. Two structural relations follow once $\theta_c = \ln \varphi$ is imposed: a width relation $w = A/\ln \varphi$ and a recursive relation between the modulation centre and the matter– Λ equality epoch.

The locked construction returns $\Delta\text{BIC} = -26.44$ on PP-excluded data ($N = 37$) and $\Delta\text{BIC} = -30.31$ on PP-included data ($N = 1738$) at $k = 2$ against ΛCDM at $k = 3$. The locks survive a $47\times$ enlargement of the data set, and locking four of six modulation parameters costs only $\Delta\chi^2 = +1.00$ relative to the unconstrained fit. The data therefore point essentially at the lock pattern within their resolving power.

The existence of an exact topological reconstruction of the observed profile is established. Whether this structure reflects a genuine cosmological selection principle for φ , or whether the agreement is a feature of a symmetry that admits multiple possible origins, remains open. The present paper does not adjudicate this question; it records the reconstruction as it stands.

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A Numerical reproducibility

All numerical values reported in this paper were produced with the χ^2 pipeline reproduced in v0.3 [3] from the v0.1 reference script [1]. The Pantheon+ data files `Pantheon+SH0ES.dat` and `Pantheon+SH0ES_STAT+SYS.cov` were obtained from the official `PantheonPlusSH0ES/DataRelease` repository on GitHub via sparse checkout. The compressed Planck 2018 distance priors and the

BAO data sets enter the PP-excluded subset ($N = 37$) and the PP-included subset ($N = 1738$) as in v0.1 and v0.3. The Lagrangian-side identities of Section 4 (Eqs. 8, 10, 11) were verified by direct symbolic substitution and by numerical evaluation at the sample points listed in the text, with absolute residuals below 10^{-16} .

The PP-included scale-up reported in Section 3 was produced by holding the locks of Eq. (4) fixed and refitting only H_0 and $\Omega_b h^2$ by differential evolution followed by Nelder–Mead polish on the full $N = 1738$ likelihood. The Λ CDM reference $\chi^2 = 1624.848$ at $k = 3$ on the same data set reproduces the v0.1 published value [1] to within 0.005.

Equation (14) was tested against the data by locking Ω_m to its right-hand side and refitting $H_0, \Omega_b h^2$ by differential evolution followed by Nelder–Mead polish; the lock incurred $\Delta\chi^2 = +0.0005$ against the free- Ω_m reference on PP-excluded data. The candidate $\Omega_m = 1/(2+\varphi)$ was tested in the same way and incurred $\Delta\chi^2 = +25.7$. The 1σ confidence interval on A reported in Section 6 was obtained by profile likelihood with Ω_m locked by Eq. (14) and the other modulation parameters tied to A by Eq. (12).

A standalone Python reproduction script is provided as an ancillary file with this submission and is also archived at Zenodo under MIT licence; see Code Availability below.

Code Availability

The Python pipeline used to produce all numerical results in this paper is provided as an ancillary file `lvc_v04_reproduce.py` with this arXiv submission and is also archived at Zenodo (DOI to be assigned) under the MIT licence. The repository contains the reproduction script, the licence file, and a README describing the data acquisition procedure for the public Pantheon+ data files.

References

- [1] LUMENPIXEL, *Lagrangian Variable Cosmology v0.1: Theoretical Reconstruction of the Lock- φ Modulation under a Phase-Rotational Spacetime-Unfolding Hypothesis*, Working Paper, Zenodo (May 2026). DOI: [10.5281/zenodo.20079027](https://doi.org/10.5281/zenodo.20079027).
- [2] LUMENPIXEL, *Lagrangian Variable Cosmology v0.2: Coordinate Stability of the Lock- φ Modulation under Synthetic-Data Self-Consistency Tests*, Working Paper, Zenodo (May 2026). DOI: [10.5281/zenodo.20079084](https://doi.org/10.5281/zenodo.20079084).
- [3] LUMENPIXEL, *Lagrangian Variable Cosmology v0.3: Application of the v0.2 Section 10 Protocol to Real Data and Diagnostic Reframing of the v0.1 Width Lock*, Working Paper, Zenodo (May 2026). DOI: [10.5281/zenodo.20079090](https://doi.org/10.5281/zenodo.20079090).
- [4] LUMENPIXEL, *LVC v16: Combined DR1+DR2 BAO Fit Identifies a New Lock Candidate (Lock- φ) at $z_c \approx 1/\varphi$* , Working Paper, Zenodo (May 2026). DOI: [10.5281/zenodo.20078901](https://doi.org/10.5281/zenodo.20078901).
- [5] DESI Collaboration, *DESI 2024 cosmological constraints from the measurement of baryon acoustic oscillations*, arXiv:2503.14738.
- [6] D. Brout *et al.*, *The Pantheon+ Analysis: Cosmological Constraints*, *Astrophys. J.* **938**, 110 (2022).
- [7] A. G. Riess *et al.*, *A Comprehensive Measurement of the Local Value of the Hubble Constant with $1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ Uncertainty from the Hubble Space Telescope and the SH0ES Team*, *Astrophys. J. Lett.* **934**, L7 (2022).
- [8] Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, *Astron. Astrophys.* **641**, A6 (2020).

- [9] S. Alam *et al.*, *The clustering of galaxies in the completed SDSS-III Baryon Oscillation Spectroscopic Survey: cosmological analysis of the DR12 galaxy sample*, Mon. Not. R. Astron. Soc. **470**, 2617 (2017).
- [10] eBOSS Collaboration, *The Completed SDSS-IV extended Baryon Oscillation Spectroscopic Survey: Cosmological Implications from two Decades of Spectroscopic Surveys at the Apache Point Observatory*, Phys. Rev. D **103**, 083533 (2021).